CYBER SECURITY & ETHICAL HACKING INTERNSHIP

TASK 1 – CYBERSECURITY FOUNDATIONS, LINUX ADMINISTRATION, NETWORKING FUNDAMENTALS & SECURE LAB ENVIRONMENT SETUP

Lab Report

Student Name	Mahadev R
Academic Background	B.Tech – Electronics and Communication Engineering
Task	Task 1
Project	Secure Network Asset Inventory & System Information Scanner
Environment	Kali Linux virtual machine / VirtualBox

1. Introduction

Task 1 establishes the technical and ethical foundation required for cybersecurity practice. The task focuses on cybersecurity fundamentals, virtualization, Linux administration, networking, Python-based security utilities, Git and GitHub, and secure laboratory configuration. The practical work is performed in an isolated environment so that security experimentation can be carried out safely and with appropriate authorization.

2. Objectives

The main objective was to build a functional cybersecurity laboratory and develop foundational skills required before moving to reconnaissance, scanning, vulnerability assessment and penetration-testing activities. The task also required development of a lightweight Python utility capable of collecting basic system and network information and saving the results as a report.

3. Cybersecurity Fundamentals

Cybersecurity involves protecting systems, networks and information from threats and unauthorized activity. Ethical hacking is performed only on systems for which permission has been provided. An important foundation is the CIA Triad: Confidentiality protects information from unauthorized disclosure, Integrity protects information from unauthorized modification, and Availability ensures authorized users can access resources when required.

4. Virtualization and Secure Lab Environment

A virtual machine provides an isolated environment in which cybersecurity exercises can be performed without directly experimenting on the host operating system. The lab architecture used for this task follows the structure: Host Operating System → VirtualBox → Kali Linux VM → Internal Virtual Network. Virtualization also supports snapshots, which provide a recovery point before significant configuration changes.

5. Kali Linux and Linux Administration

Kali Linux was used as the practical Linux environment. Terminal-based administration was practiced using commands for system information, network configuration, directory navigation, file listing and Python execution. Linux file-system concepts were also documented, including /home for user files, /etc for configuration, /var for variable data such as logs, /usr for programs and libraries, and /bin for essential commands.

6. Networking Fundamentals

The task covered the OSI and TCP/IP models, IPv4 addressing, subnet basics, MAC addresses, routing, switching, DNS, DHCP, ports and common protocols. The practical environment used virtual networking to support controlled communication between lab systems. Common protocols considered include HTTP, HTTPS, DNS, DHCP, SSH, FTP and ICMP. Understanding these concepts is essential for interpreting network traffic and performing authorized security assessments.

7. Secure Network Asset Inventory & System Information Scanner

The mini project is a Python command-line utility designed to collect basic information about the local system and network. Its expected functions include hostname detection, local IP address detection, MAC address retrieval, operating-system information, network-interface listing and report generation. The project produces a text report containing the collected information.

8. Project Implementation

The main source file is scanner.py. The application is executed from the Linux terminal using Python 3. The resulting system information is saved in system_report.txt. The project demonstrates the use of Python for basic security-oriented system inventory and reinforces command-line, operating-system and networking concepts.

9. Git and GitHub

The completed project was organized in a public GitHub repository named cybersecurity-task-1. The repository contains the Python source file, generated system report, README documentation, screenshots and diagrams. GitHub provides version-control and project-documentation capabilities and makes the completed work available as a structured technical submission.

10. Evidence and Documentation

Practical evidence was collected through screenshots showing the virtual environment, Kali Linux, Linux commands, network information and execution of the Python scanner. Supporting diagrams were prepared for the cybersecurity lab architecture, OSI model, TCP/IP networking and Linux file-system hierarchy. These materials document both the configuration and the concepts covered during Task 1.

11. Learning Outcomes

Completion of this task provided practical exposure to cybersecurity fundamentals, Linux administration, virtualization, networking, Python security utilities, Git/GitHub and technical documentation. These foundations support the later stages of the internship, including reconnaissance, scanning, vulnerability assessment and penetration-testing activities.

12. Conclusion

Task 1 established a controlled cybersecurity laboratory and connected foundational theory with practical implementation. The successful completion of the Secure Network Asset Inventory & System Information Scanner demonstrates basic automation of system and network information collection. The documented lab, diagrams, screenshots and GitHub repository provide a structured foundation for subsequent cybersecurity tasks.

Appendix – Submission Contents

Item	Status / Location
Python source code	scanner.py
Generated report	system_report.txt
Project documentation	README.md
Screenshots	GitHub screenshots folder
Diagrams	GitHub diagrams folder
GitHub repository	cybersecurity-task-1